

Known Truth, Ambiguous Measurement: A Controlled Study of Grammar Knowledge Tracing

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Abstract

Language-learning logs rarely expose trustworthy ground truth for grammatical opportunity boundaries, knowledge components (KCs), item–KC mappings, or learner mastery. Synthetic data can expose those quantities, but known latent truth does not make the observable measurement valid. We build and preserve a clean grammar-KT benchmark from all 1,222 records of a grammar resource: 75 canonical constructions, 18 declared generator factors, 113 prompts, a full-rank Q-matrix, and 283,000 interactions from 1,000 synthetic learners. Its known truth enables counterfactual studies unavailable in ordinary logs: KC coarsening/refinement, Q misspecification, ontology equivalence, state recovery, and collection geometry. We then audit the measurement process rather than assuming it is realistic. Two model-based item audits agree that only 60/113 prompts are usable as stored and flag 18 critical answer-space or withhold cases; three outcome-blind KC inductions share only nine activation signatures. A preregistered four-format bank construction passes all gates for only 5/38 matched families, so we release no “realistic” successor dataset. In a separate content-free structural control, planted format nuisance makes a false format-specific KC split relatively more predictive (three-seed difference-in-differences $-.0316$); an explicit format covariate outperforms that split. Oracle-linked structured errors greatly improve failed-KC localisation but yield only small next-response gains. A dialogue continuum improves visible naturalness while sharply reducing answer determinacy and KC attribution. The resulting contribution is not a synthetic human clone: it is an auditable analysis of how KC structure, measurement nuisance, response information, and assignment policy determine what grammar KT can infer.

1 Introduction

Knowledge tracing (KT) estimates changing learner knowledge from chronological responses (Corbett and Anderson, 1995; Abdelrahman et al., 2023). Its input is usually treated as a sequence of items, outcomes, and knowledge-component (KC) labels. Yet the label matrix embeds a measurement theory: it declares what knowledge is reusable, what an item requires, and where practice should transfer. A large response log cannot by itself reveal whether those declarations are valid.

This problem is acute in language learning. In a dialogue corpus such as the Teacher–Student Classroom Corpus v2, a turn may be explanation, elicitation, repair, clarification, or several grammatical phenomena rather than one scorable opportunity (Caines et al., 2022). Large platform logs record behaviour selected by a curriculum or adaptive policy, while open responses admit alternative correct forms and structured errors. The true opportunity boundary, KC ontology, Q-matrix, and learner state remain latent. Consequently, real logs cannot directly answer whether a recovered ontology is the generating one, whether a KC split is absorbing item difficulty, or what would have happened under a counterfactual measurement policy.

We study a perfect-information thought experiment. What analyses would be possible if a language-learning dataset supplied true grammatical opportunities, generator KCs, Q, and learner mastery? Because those quantities are prohibitively difficult or impossible to observe jointly in human data, we construct them synthetically. The synthetic world provides experimental truth—not human cognitive truth.

Perfect annotation creates internal validity but does not solve measurement. If a learner cannot infer what response is requested, if several reasonable answers are marked wrong, or if a false KC split captures a task-format effect, then a mathe-

matically convenient benchmark can still mislead. We therefore separate five layers:

$$\begin{aligned} \text{GRAMMARCELL} &\rightarrow (K^*, Q^*) \rightarrow \text{instrument} \\ \text{instrument} &\rightarrow \text{policy} + \text{learner} \rightarrow (\hat{K}, \hat{Q}). \end{aligned} \quad (1)$$

A GRAMMARCELL is a canonical construction; K^* is a declared latent coordinate system possessed by synthetic learners; and $\hat{K}$ is a downstream hypothesis. Thus $\text{GRAMMARCELL} \neq K^* \neq \hat{K}$.

Our immutable reference, `grammar_kt_full_v1`, processes every English Grammar Profile record under an explicit verbal-morphosyntax scope (O’Keeffe and Mark, 2017). It freezes 75 cells, 18 generator factors, 113 prompts, a rank-18 $113 \times 18 Q^*$, and 283,000 binary events before any downstream experiment. It is a clean Q-driven laboratory environment: prompt text is auditable, but the response simulator consumes only Q and mastery.

We ask three questions:

- RQ1 What must be declared and audited to construct a grammar-learning measurement environment with known latent structure, and where does a clean Q-driven benchmark fall short?
- RQ2 How do KC/Q misspecification and item- or format-specific nuisance affect prediction and recovery of the planted learner state?
- RQ3 Which item-bank contrasts, assignment histories, and response information make latent structure and mastery distinguishable?

Our contributions are fourfold. First, we preserve a replayable clean-control benchmark whose oracle truth supports counterfactual representation and identifiability studies unavailable in real logs. Second, we conduct a complete learner/platform audit, a pedagogical KC audit, and repeated outcome-blind KC induction; these expose substantive validity limits that Q rank alone cannot answer. Third, we attempt a full-rank matched-format bank and retain its failure—only 5/38 complete families pass—rather than silently relaxing the release gate. Fourth, we isolate planted format/item nuisance, learner heterogeneity, structured errors, assignment policies, and dialogue openness in separate controlled studies. The result is a methodology for asking when KT is informative or misleading. It does not claim a deployable platform dataset,

human-like simulator, or uniquely correct grammar ontology.

2 Related Work

Grammar resources and opportunity boundaries. The CEFR provides proficiency scales, while the English Grammar Profile (EGP) links corpus-supported grammar competence statements to CEFR levels (Council of Europe, 2020; O’Keeffe and Mark, 2017). These are rich linguistic resources, not item banks or cognitive models. TSCCv2 retains discourse relations, pedagogical focus, repair, and correction in real English tutoring (Caines et al., 2022); dialogue KT must infer which turns are opportunities and what each tests (Scarlato et al., 2025). We study the complementary setting in which opportunity and generator truth are declared, then ask how much ecological context can be added before that boundary becomes ambiguous.

KC/Q specification and identifiability. KCs connect performance to instructional units (Koedinger et al., 2012). Classical BKT models learning, guess, and slip (Corbett and Anderson, 1995); PFA uses logistic practice features (Pavlik et al., 2009). Cognitive-model refinement combines statistical evidence with expert interpretation (Cen et al., 2006), and response data can validate or refine Q (de la Torre and Chiu, 2016; Pardos and Dadu, 2018). Q misspecification can be absorbed into response parameters and damage latent classification (Rupp and Templin, 2008). Formal Q-identifiability conditions are response-model-specific; for example, DINA results require structure beyond numerical rank (Gu and Xu, 2021). We therefore separate linear Q geometry, empirical recovery, pedagogical interpretation, and psychological truth.

Measurement validity, item difficulty, and format. Validity concerns the interpretation and use of responses, including construct underrepresentation and construct-irrelevant variance (Messick, 1994; Kane, 2013). Item difficulty is central in IRT and has been incorporated into KT (Pardos and Heffernan, 2011); individualized KT shows that learner learning rates can also matter (Yudelson et al., 2013). Format effects are heterogeneous: matched multiple-choice and constructed-response stems are more comparable than unmatched content, and language-test effects depend on design

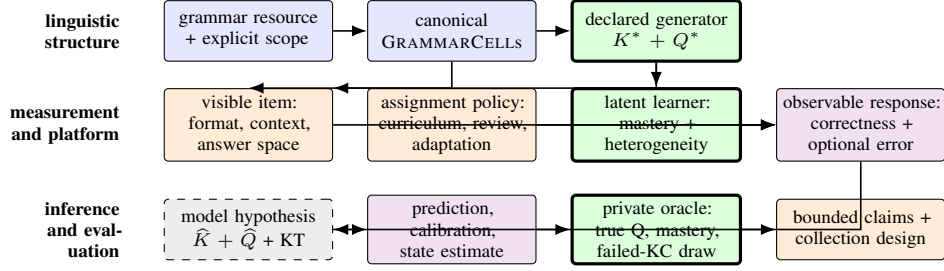


Figure 1: Known latent truth and observable measurement are different layers. Responses arise jointly from learner state, the visible instrument, and the platform policy. Model hypotheses consume observable histories; private oracle fields are opened only for declared evaluation. Full-v1 fixes the green truth but idealises the orange measurement layer.

and proficiency (Rodriguez, 2003; In’nami and Koizumi, 2009). Thus a format contrast requires matched semantic content and crossed KCs, not unrelated item campaigns. We use zero and planted controls rather than assign one supposedly universal difficulty ordering to task types.

Platform selection and learner responses. SLAM includes millions of language-production tokens with exercise, time, and error metadata (Settles et al., 2018); EdNet mixes problem, lecture, purchase, and within-question events at much larger scale (Choi et al., 2020). Such logs are selected service histories, not random item assignment. Spaced-practice models make elapsed exposure part of the learning process (Settles and Meeder, 2016). We compare a laboratory schedule with simple curriculum, mixed, and observable-history adaptive alternatives, while explicitly avoiding a policy-value claim.

Binary correctness also discards linguistic evidence. Learner corpora and ERRANT show that grammatical errors can be typed, while multiple valid corrections complicate exact scoring (Bryant et al., 2017, 2019). We begin with structured categories and linked/shuffled information controls; we do not call generated labels human errors or generate surface text without validation.

Synthetic learner studies. Known truth makes controlled KT evaluation possible (Piech et al., 2015; Pagonis et al., 2024), but simulator validity is purpose-specific, and educational simulations often provide limited validity evidence (Kaser and Alexandron, 2024). Our response is not one increasingly ornate “realistic” simulator. We retain a clean world and add separate paired worlds for item, format, learner, error, and policy mechanisms, each with a negative or positive control and a bounded

claim.

3 Scientific Objects and Source Data

3.1 Canonical grammar cells

The empirical scope is single-main-clause English verbal morphosyntax. For source descriptor d , constrained normalisation returns a status, one or more branches, explicit uncertainty, Phase-2 eligibility, and provenance. A scalar GRAMMARCELL

$$c = (t, a, v, p, q, m) \quad (2)$$

contains tense, aspect, voice, polarity, clause type, and modal choice. Table 1 gives the domains. The value NA is a licensed structural value, not missing evidence. Compatibility rules include $m \neq \text{none} \Rightarrow t = \text{NA}$ and imperative clauses requiring $t = \text{NA}$ and $m = \text{none}$.

Phase 1 receives typed descriptor fields but no examples. It may return complete scalar branches, bounded alternatives, explicit missing constraints, or a zero-cell status. Phase 2 is called only for an eligible uncertain dimension with a licensed example; it may narrow named uncertainty but must preserve every Phase-1 branch and every ineligible exact value. Deterministic validation checks domains, statuses, compatibility, branch preservation, and resume-input identity. Only complete scalar mappings canonicalise. A separate relation table retains every descriptor-to-cell link, so repeated source evidence does not become a grammatical feature.

3.2 Complete source census

The consult-only input is a parsed 1,222-record EGP snapshot with a declared SHA-256 digest. We process every row rather than selecting descriptors likely to fit the schema. Descriptions outside the declared verbal-morphosyntax boundary remain

Dimension	Symbol	Closed scalar domain	Principal constraint or interpretation
Tense	t	present, past, NA	NA is structural and differs from missing evidence
Aspect	a	none, progressive, perfect, perfect–progressive	Encodes auxiliary-chain aspect
Voice	v	active, passive	Passive is canonical BE-passive only
Polarity	p	positive, negative	Verbal polarity, not lexical meaning
Clause	q	declarative, polar question, subject-WH, non-subject-WH, imperative	Question subtypes and imperatives are distinct
Modal	m	none; can, could, may, might, must, shall, should, will, would	A central modal requires $t = \text{NA}$

Table 1: The six dimensions of a scalar GRAMMARCELL. During partial normalization, a dimension may instead hold a finite list or null; only scalar cells enter canonicalization.

auditable out_of_scope rows. This is full processing of the source under a bounded grammar scope, not coverage of all phenomena described by EGP.

The final census contains 211 complete, 327 partial, nine unresolved, and 675 out-of-scope mappings. Complete mappings yield 228 source–cell relations and 75 unique cells. A fresh balanced repeat of 120 Phase-1 rows gives 93.3% status agreement and 95.8% eligibility agreement. All 38 rows complete in both annotations have the same canonical cell set; partial-branch agreement is 79.0%. This supports explicit uncertainty rather than silent completion, but does not constitute expert annotation.

3.3 Semantic grammar regimes

Before simulation, a deterministic structural rule assigns 54 cells to seen, 15 to unseen_combination, and six to unseen_value. Each unseen combination contains only values and lower-order value pairs found in seen grammar, while its complete six-tuple is absent. The unseen-value cells contain the withheld perfect-progressive aspect value. All generator KCs nevertheless have seen support: this cohort tests a new linguistic composition, not a latent KC observed only at evaluation. Acquisition uses seen grammar only; every item receives one terminal, non-updating probe.

3.4 Clean observable and private layers

The frozen public record contains learner ID, item ID, sequence, correctness, phase, pass, and grammar regime. Generator KCs, mastery before/after, response probability, and random draw are private oracle fields. This matches the perfect-information thought experiment: ordinary KT receives only

fields a platform could log, while evaluation may later open otherwise unavailable truth.

The surface and response process are less realistic. The 113 prompts do not have an executable UI/scorer, and simulated learners never read or answer their text. Full-v1 therefore has strong artifact and experimental validity but unestablished response-process validity. We retain it unchanged and assess the measurement layer separately rather than reinterpret binary events as human production.

4 Known-Truth Construction and Measurement Programme

4.1 Outcome-independent generator KCs

Generator K^* is declared after canonicalisation and before item or learner outcomes exist. The selected reusable-operation hybrid contains 18 KCs: present and past finite-form selection; shared perfect and progressive dependencies; canonical BE-passive and verbal negation; imperative, polar-question, and non-subject-WH operations; and one KC for each of nine central modals. Active, positive, declarative, simple, modal-free, tenseless, and unsupported subject-WH conditions are not separate latent states. Perfect-progressive activates both shared aspect operations.

The declaration is stored as machine-readable activation rules. Generic projection code contains no English KC names. Before item construction, we also audited a 19-KC feature-only control, a hybrid with a perfect-progressive-chain KC, and a 75-KC exact-cell upper bound. The chain was rejected because its six-cell column is strictly nested in both components and adds no independently motivated operation; exact cells have no reuse. These

decisions use linguistic coherence, reuse, support geometry, and parsimony, never response prediction.

4.2 Measurement-bank construction

For each of 75 cells, the default campaign draws three independent contextual controlled-production candidates from one model configuration. Generation receives the cell, English rulebook, item-format declaration, and a common-language instruction, but no learner outcomes, regime labels, KT metrics, or discovered KCs. A deterministic answer/slot check precedes an independent model validator. Acceptance requires target fidelity, grammaticality, naturalness, pedagogical suitability, determinacy, simple non-target language, no answer leakage, no extraneous grammar, and no unnecessary world knowledge.

Raw evidence is immutable. The default campaign left 18 cells uncovered, so we executed a frozen two-draw unchanged-prompt rescue, followed by a separately labelled explicit-construction intervention for the remaining determinacy-dominated cohort. An append-only accepted-answer correction fixed one finite omission. Open full-sentence imperative production remained indeterminate; a final preregistered all-and-only unordered-cue format bounded the response space while retaining imperative word order and negative DO-support. No repair is hidden as default generation.

Outcome-free curation retains at most two distinct accepted prompts per cell: the first and, where available, a contextually distant second candidate. This produces 113 unique prompts: 37 cells have one item and 38 have two. Max-one would retain 75 items; up-to-three adds 13 variants but no cell. The chosen bank supports contextual replication without treating surface variation as new grammatical structure.

Prompt and accepted-answer text are central to generation, validation, and the intended learner-facing measurement surface, but they have no causal role in the baseline response generator after the bank is frozen. The simulator neither renders a prompt nor collects or scores free text: an item identifier selects its Q^* row, mastery over the active KCs determines response probability, and keyed randomness samples binary y . Table 10 shows how the same stored surface can pass linguistic checks yet fail answerability or KC purity. The baseline is consequently a controlled forward-generation and

inverse-recovery benchmark, not a simulation of linguistic production.

4.3 True Q-matrix and Q-geometry gate

For fixed item i and generator KC k ,

$$Q_{ik}^* = \mathbb{I}[k \text{ activates on } c_i]. \quad (3)$$

Projection is deterministic from the item cell and frozen KC rule; response data never enter. Before simulation, a mandatory gate measures item and cell support, row widths, KC co-occurrence, isolated opportunities, identical and near-identical columns, activation-equivalence classes, and matrix rank.

The final 113×18 Q^* has 269 edges, density .1323, rank 18, 75 distinct cell-activation rows, no identical columns, and no column pair with Jaccard similarity at least .90. Each KC has at least two items, although seven have fewer than six. Forty-six KC pairs have A-only, B-only, and A+B evidence; 105 have two-sided evidence without co-occurrence. The sole non-subject-WH cell leaves two nested pair geometries, retained as a source inventory limitation.

4.4 Baseline learner simulator

The simulator consumes only fixed items, K^* , Q^* , a protocol, and a seed. Learner-by-KC initial mastery is independently Beta(2, 2). For an item with active set A_i , the response probability is

$$p(y_i = 1) = .10 + .80 \min_{k \in A_i} m_k. \quad (4)$$

Every active KC receives outcome-independent opportunity learning $m'_k = m_k + .02(1 - m_k)$. The baseline has no forgetting, item difficulty, prerequisites, or transfer. Minimum aggregation was frozen because it is permutation invariant, noncompensatory, and invariant to repeating an equal-mastery prerequisite; product and compensatory means fail at least one of these interpretation checks.

A seen-only Q^* -balanced schedule gives each learner at least 12 opportunities per seen KC. It contains 170 acquisition events, followed by one non-updating probe of all 113 items. For 1,000 learners this yields 283,000 rows. Public events contain only learner, item, sequence, correctness, phase, pass, and grammar-regime fields. Active generator KCs, mastery before/after, response probability, and random draw are stored separately as private oracle truth.

4.5 Frozen release and downstream hypotheses

The versioned release records commands, configurations, hashes, deterministic gzip settings, and a recursive artifact inventory. An independent replay verifies every public/private link and state transition. Downstream studies may hide or perturb K^*/Q^* and construct $\hat{K}/\hat{Q}$, but cannot mutate the release. Pure representation comparisons reuse the same learner events. Oracle fields are opened only for explicitly labelled post-selection structural or mastery evaluation.

4.6 Validity audit beyond Q geometry

We next assess internal and ecological validity separately. A complete strict census codes comprehension, answer determinacy, pedagogical and format plausibility, lexical/context burden, response-space fairness, KC purity, shortcuts, and deployability. A second frozen audit obtains learner, teacher, platform-product, and measurement judgments independently. We retain exact item IDs, rationales, role disagreements, and a declared cross-audit label map; no scalar “realism” score overrides a critical answer-space failure. These model judgments are automated stress tests, not human evidence.

We audit K^* separately for support, isolates, nesting, plausible independent mastery, pedagogical transfer, and format/lexical confounds. To measure dependence on one decomposition, three independent outcome-blind KC inductions receive only frozen cells, support information, and an activation predicate grammar. Proposals are canonicalised by which cells they activate, not by wording. Generator truth remains unchanged regardless of their output.

4.7 Matched-format construction and release gate

Twelve cell-Q rows cannot have rank 18, so we select 18 seen cells with a rank-18 witness, plus one unseen-combination and one unseen-value probe cell, before response outcomes. The target bank crosses each seen cell with four formats (constrained cloze, dialogue completion, multiple choice, and sentence transformation) and two semantic variants; held-out cells receive one variant. Each family shares a semantic specification and Q row.

Three append-only generation rounds pass through deterministic schema/target checks, inde-

pendent learner solvers, and linguistic, measurement, and product critics. A family passes only if all four formats pass; raw attempts are never silently repaired. Exact crossing, answerability, measurement purity, cell/KC coverage, held-out coverage, and seen-Q rank are hard release gates. Failure means no measurement dataset, not a lower-quality partial release.

4.8 Content-free nuisance worlds

Because the learner-facing bank fails (Section 6), we do not simulate it. Instead, a separate non-release instrument instantiates only the preselected 152-slot geometry and categorical format labels. It contains no prompt, answer, or scorer. For minimum active mastery m , responses use

$$p = g_l + (1 - g_l - s_l) \sigma[\text{logit}(m) + a_l + \delta_f - d_i], \quad (5)$$

where a_l , δ_f , and d_i are learner, format, and item effects. Six paired worlds separately set these effects to zero; add moderate/strong format offsets; add moderate item difficulty; cross item and format; and add learner ability, learning-rate, guess, and slip heterogeneity. Item effects are orthogonal to intercept, Q, and format over equally weighted seen slots. All effect sizes are planted stress levels, not human estimates.

Each world uses 500 learners, seeds 20260829–20260831, 188 acquisition events, and 152 non-updating probes. Keyed common random numbers pair disabled and enabled mechanisms. Public and oracle streams remain separate. Four observable models compare shared K^* without nuisance (A), false format-split KCs (B), shared K^* plus format (C), and shared K^* plus format and a seen-item residual basis (D). D deliberately spans the planted seen-item effect and is a positive control, not a general difficulty solution.

4.9 Policy, errors, and dialogue

The Q-balanced laboratory schedule is compared under the same budget with a fixed curriculum order, mixed practice, and an adaptive policy using only observable cell-level correctness, exposure, and spacing. The latter cannot read generator Q or mastery. Policy model fits are declared post-response exploratory; the confirmatory comparison is process morphology.

For incorrect multi-KC responses, a private diagnostic KC is sampled proportional to active mastery deficits. It is post-outcome attribution, not a unique

causal failure. We expose binary-only, oracle-linked positive-control, 80%-linked/unresolved, and within-item shuffled error histories while holding all other observable fields fixed. Current-row errors update history only after prediction.

Finally, four visible item families span cloze, transformation, contextual production, dialogue completion, and open dialogue. Independent role critics judge naturalness, determinacy, response-family size, incidental grammar, shortcuts, KC attribution, and platform plausibility separately. This is an automated opportunity-boundary stress test, not learner validation.

5 Experimental Design

5.1 Clean-control representation evidence

The immutable full-v1 stream supports the known-truth baseline. A standardized PFA-like logistic model uses strictly prior successes and opportunities on active hypothesis KCs (Pavlik et al., 2009). It receives no item identity, generator state, response probability, random draw, or probe outcome. Six deterministic representations span one all-merged KC, six family unions, K^* , 35 split-2 KCs, 66 split-4 KCs, and 75 exact cells. Nine frozen Q-corruption controls add false-positive, false-negative, or mixed edges at a 27-edge budget. All representations reuse the same 113,000 probes.

Observable-only discovery searches 181 outcome-independent features, operations, unions, splits, supported intersections, exact cells, and hash distractors using learner-disjoint fit/validation groups. It cannot read K^* , Q^* , oracle state, or probes before selection. Name-free activation matching opens truth only after the representation is frozen. Separate collection controls vary learner count, opportunities, within-cell replication, and A-only/B-only/A+B anchors. These retained experiments answer whether response volume or Q geometry distinguishes hypotheses; they do not validate the learner-facing instrument.

5.2 Audit and construction outcomes

The two item audits cover all 113 full-v1 prompts. The strict census is one rubric-guided review; the second comprises four role-specific fixed calls per item. A deterministic script maps their labels, reports exact disagreement, and preserves item IDs for every cell. The KC audit is outcome-free. Three KC inductions use independent calls and are compared by activation signatures.

The matched-bank campaign treats a complete cell/variant family as the unit: all four formats must pass deterministic checks, independent solver coverage, and three role critics. Three declared rounds may generate new candidates but may not edit prior output. The primary outcome is the number of complete families and resulting cell/KC support and Q rank; isolated passing slots cannot satisfy the release gate.

5.3 A–D nuisance comparison

The controlled instrument contains 144 seen and eight held-out structural slots. For each of six worlds and three seeds, 500 learners generate 170,000 events under a Q-balanced schedule. Nine additional response runs use the combined heterogeneous world with curriculum, mixed, or adaptive assignment, for 4.59 million events overall.

Learners are split 60/20/20 by a stable hash. Model selection uses training learners’ acquisition rows and dev learners’ seen probes; the selected penalty is refit on train+dev acquisition and evaluated on test learners’ terminal probes. Standardization is fit on training acquisition only. Primary scope is seen-probe log loss; Brier score, fixed-width ECE, AUC, and an oracle-only item-prerequisite state RMSE are secondary. Model D’s item basis covers seen item IDs only; held-out probe items are zero encoded.

The four preregistered contrasts are:

$$\begin{aligned} &[(L_B - L_A)_{\text{strong format}} - (L_B - L_A)_{\text{zero}}], \\ &(L_C - L_B)_{\text{strong format}}, \\ &(L_D - L_C)_{\text{item only}}, \quad (L_D - L_C)_{\text{item+format}}. \end{aligned}$$

Negative values in the first contrast mean format nuisance increases B’s relative advantage over A; they do not validate the false split. The remaining negative values favour the explicitly adjusted candidate.

5.4 Error, policy, and dialogue evidence

Error-history models hold D fixed and vary only prior observable error labels. We evaluate prediction, item-prerequisite state, failed-KC localisation on incorrect multi-KC probes, and a transparent Beta-smoothed terminal-KC evidence diagnostic. The latter is not a fitted KT state. The shuffled stream preserves within-item/phase category marginals and is a negative learner-specific information control, not a claim of complete statistical independence.

Schedule diagnostics report acquisition and probe accuracy separately, item/KC exposure inequality, repetition gap, and adjacency. Q-balanced, curriculum, and mixed schedules share the same occurrence multiset; under unconditional order-independent learning they have identical terminal oracle mastery by construction. Alternative-policy A–D fits were frozen after responses and descriptive diagnostics, so they are exploratory.

The dialogue study generates four matched families at five openness levels and collects five role judgments for each of 20 opportunities. We report each dimension and disagreement; no composite is computed.

5.5 Uncertainty and model assistance

Within a frozen fitted model, percentile intervals re-sample 82 test learners 2,000 times. They condition on the fit, selected hyperparameter, bank, seed, and simulator world; they omit refitting, item/cell sampling, simulator-family uncertainty, and multiplicity. Three-seed means and ranges are descriptive, not population intervals.

Retained construction and audits use fixed gpt-5.6-sol or gpt-5.6-terra calls with prompt, input, schema, settings, raw output, parsed output, and hashes frozen. Model critics are independent stress tests, not human participants. No learner, teacher, or platform expert judgment is fabricated.

6 Results

6.1 RQ1: auditability does not imply platform validity

All 1,222 source rows receive a disposition. The complete in-scope subset yields 75 canonical cells; the outcome-independent release freezes 18 generator factors, 113 prompts, a 113×18 Q with 269 edges and rank 18, and 283,000 replayable events. This establishes artifact integrity and a usable perfect-information control.

The stronger measurement claim fails. In the strict prompt census, 70/113 items are usable under an assumed response-slot UI, 15 need minor repair, 15 are answerable but artificial, 10 have a material answer-space problem, and three require rewrite/withholding. A four-role audit yields 452 judgments. After a declared mapping, the audits agree exactly on 70 items; only 60 are usable in both, 53 occur in the union requiring action, and 18 in the critical answer-space/withhold union. The

four roles disagree internally on 56 items. These are model-based warnings, not estimates of human deployability.

The structural KC audit likewise narrows the generator claim. Only 16/113 items are single-KC rows and six KCs lack an isolating opportunity. Three outcome-blind KC inductions produce 17/18/18 unique activation signatures with ranks 17/18/17. Only nine signatures are common to all three, their union has 30, pairwise Jaccard is .400–.458, and 5/4/7 exactly match K^* . The 18 columns remain a declared, useful experimental coordinate system, not a uniquely recovered human ontology.

The matched-format construction makes the measurement gap concrete. Across three rounds, all 178 calls complete technically, producing 106 candidates, 712 solver attempts, and 90 role judgments. The family funnel is $38 \rightarrow 31 \rightarrow 12 \rightarrow 3$ in round 1, $35 \rightarrow 32 \rightarrow 12 \rightarrow 2$ in round 2, and $33 \rightarrow 26 \rightarrow 6 \rightarrow 0$ in round 3 (deterministic, solver, critic gates). Only 5/38 complete families (20/152 slots) pass, covering 4/20 cells, 6/18 KCs, and seen-Q rank 3. Measurement critics accept 9/30 critic-reached candidates versus 29/30 linguistic and 24/30 product decisions; 23/30 decisions are mixed. No new dataset is released.

6.2 RQ2: representation and measurement nuisance interact

In clean full-v1, all-probe loss is minimized at K^* (.670627). Family merging costs .008132, split-2 .003165, split-4 .005868, and exact cells .015039; every learner-paired interval excludes zero. All nine frozen 10% Q-corruption instances are harmful. Oracle prerequisite-state RMSE agrees overall: .123738 for K^* versus .146300 family-coarse, .132752 split-2, .140428 split-4, and .163828 exact cells. These are recoverability results in the world generated from K^* , not evidence that K^* is human truth.

Legacy simulator controls already expose one important exception: under unmodelled item logit difficulty, split-2 beats K^* in one of three seeds. The matched structural experiment isolates that alternative explanation.

The zero-format negative control behaves correctly: A/B mean loss is .657219/.664138, so an unnecessary format split hurts. Under the strong planted format effect, A/B/C loss is .663597/.638965/.633648. Thus B can win by absorbing measurement nuisance, but the observed format model C performs better. The DiD is nega-

Final source disposition	Count	Share
Complete	211	17.3%
Partial	327	26.8%
Unresolved	9	0.7%
Out of scope	675	55.2%
Source records	1,222	100%
Source-cell relations	228	—
Canonical cells	75	—

Item cohort	Candidates	Accepted	Cells added/covered
Default $N = 3$	225	102	57 covered
Unchanged rescue	36	10	9 added
Explicit construction	18	9	6 added
Answer-package copies	3	1	1 added
Cue-bounded imperatives	4	4	2 added
Accepted pool	—	126	75 covered
Fixed max-two bank	—	113	75 covered

Table 2: Full-source normalisation (left) and append-only item construction (right). Each intervention was frozen and separately labelled before its calls. Accepted denotes automatic independent validation, not human expert judgment.

Automated full-v1 item audit	Count	Share
Usable under stated minimal UI	70	61.9%
Minor UI/context change	15	13.3%
Valid but pedagogically artificial	15	13.3%
Material answer-space problem	10	8.8%
Rewrite or withhold	3	2.7%
Exact cross-audit disposition	70	61.9%
Usable in both audits	60	53.1%
Either audit requires action	53	46.9%
Either audit flags critical risk	18	15.9%
Live four-role disagreement	56	49.6%

Matched-family construction	Determ.	Solver	Critics
Round 1 (38 active families)	31	12	3
Round 2 (35 active families)	32	12	2
Round 3 (33 active families)	26	6	0
Freeze requirement		Required	Retained
Whole families		38	5
Four-format item slots		152	20
Selected cells		20	4
Generator KCs covered		18	6
Seen-cell Q rank		18	3
Dataset freeze permitted		yes	no

Table 3: Learner/platform audit and matched-bank construction. The left panel is a 113-item automated census; cross-audit rows compare its strict disposition with a separate four-role model audit (452 judgments). “Critical” denotes answer-space or withhold risk. Right-panel gates are sequential within each round. Passing families are examples, not a partial release; model judgments are not human deployability evidence.

tive in all seeds (mean $-.031551$), and C-B is also negative in all seeds (mean $-.005317$).

The result is not universal. In the item-only world, B-A is $+.001195$, $+.000847$, and $-.001328$; every conditional interval crosses zero, so this experiment does not show format splits absorbing item difficulty. In the combined heterogeneous world, C-B is $-.002226$, $+.000169$, and $+.001074$, again with every interval crossing zero. D-C remains negative, but D is deliberately aligned to the planted seen-item residual span. It establishes a positive control, not arbitrary or unseen-item deconfounding.

6.3 RQ3: information and collection determine distinguishability

The clean observable selector recovers a high-overlap activation-equivalence class, not a unique ontology. Atomic and compositional policies have identical seen Q; the compositional candidate-space ceiling recovers 18/18 generator columns, while the selected atomic projection recovers 16/18 (activation Jaccard $.970854$; aligned Q-edge F1 $.965385$). Their seen predictions are identical. Hash distractors recover 0/18 and cost $.013238$,

proving the selector is informative without making equivalent rules identifiable.

Collection controls explain why. Repeating only A+B opportunities leaves factorised, union, interaction, and missing-interaction representations exactly tied through $N = 1000$. A-only and B-only anchors restore rank and make the union detectably wrong, yet a weak planted interaction remains difficult: at $N = 1000$, omitting it costs only $.000506$ in the balanced design. More responses cannot repair missing activation contrasts; numerical rank alone does not guarantee practically unique recovery.

Structured errors supply a different kind of contrast.

The linked positive control reduces mean next-response loss by only $.000867$; one of three conditional intervals crosses zero. The 80%-linked mean gain is $.000527$ and excludes zero in one seed. Nevertheless, failed-KC top-1 rises from $.421$ under binary evidence to $1.000/.884$ under exact/80%-linked evidence. The shuffled control does not improve prediction and localises poorly, but it also improves the secondary evidence-count RMSE over binary attribution by distributing multi-

Supplied representation	KCs	Probe log loss / Δ vs. K^* [95% CI]	State RMSE	Interpretation
All merged	1	+ .010225 [.009380, .011029]	–	maximal coarsening
Linguistic families	6	+ .008132 [.007450, .008779]	.146300	union merge
Generator K^*	18	.670627 (reference)	.123738	planted truth
Structural split-2	35	+ .003165 [.002744, .003581]	.132752	context refinement
Structural split-4	66	+ .005868 [.005281, .006473]	.140428	finer refinement
Exact cell	75	+ .015039 [.014108, .015990]	.163828	no reuse

Table 4: RQ2 response prediction and oracle-only prerequisite-state recovery. Log-loss intervals resample 1,000 learners 2,000 times. State RMSE compares the inverse-linked public prediction with the oracle minimum active-KC mastery on 113,000 probes; all-merged was not in the frozen mastery subset.

Controlled world	A	B	C	D	Paired log-loss contrast
Clean zero	.6572	.6641	.6572	.6578	
	.1156	.1358	.1156	.1157	$B - A = +.006919$ (3/3)
Strong format	.6636	.6390	.6336	.6341	
	.1225	.2019	.1185	.1186	$C - B = -.005317$ (3/3)
Item only	.6614	.6616	.6615	.6484	
	.1203	.1545	.1203	.1174	$D - C = -.013099$ (3/3)
Item + format	.6625	.6545	.6545	.6419	
	.1220	.1715	.1209	.1178	$D - C = -.012609$ (3/3)
Combined heterogeneous	.6561	.6487	.6484	.6364	
	.1316	.1772	.1311	.1297	$C - B = -.000327$ (0/3); $D - C = -.012031$ (3/3)
Format DiD $(B - A)_{\text{strong}} - (B - A)_{\text{zero}}$					$-.031551$ [–.033431, –.029013] (3/3)

Table 5: A–D results on 11,808 seen terminal probes per seed. Model cells stack three-seed mean log loss over item-prerequisite state RMSE (lower is better). A uses shared K^* , B a false format split, C adds observed format, and D adds a residual basis aligned to planted effects of the same seen items. Parentheses count seed-conditional learner intervals excluding zero in the stated direction. A negative DiD means nuisance increases B’s relative advantage; it does not validate B.

KC failures. No single state metric therefore proves learner-specific error information.

Policy changes the histories a model sees. Curriculum-minus-lab D loss averages +.003420 (two of three seed-conditional intervals exclude zero), mixed practice is approximately null (–.000018; all cross zero), and adaptive-minus-lab averages +.003118 (one excludes zero). Lab, curriculum, and mixed have the same occurrence multiset and exactly the same within-seed terminal mastery and probe accuracy. Their differences are history-recovery effects, not learning efficacy. Adaptive assignment changes exposure and state, so its difference mixes selection, practice, outcome, and fit.

Finally, the visible dialogue continuum exposes an ecology–precision tradeoff. Cloze receives 17/20 naturalness, 17/20 determinacy, and 17/20 clear-KC judgments, with one shortcut and mean response-family lower bound 1.30. Open dialogue receives 20/20 naturalness but 0/20 determinate judgments (16 bounded, three materially ambiguous, one not applicable), only 4/20 clear-KC judgments, 13 shortcuts, and mean response-family

bound 4.55. Open-minus-cloze matched risk is +1.00 for determinacy, +.70 for KC attribution, +1.50 incidental grammar, and +.60 shortcuts; response families grow by 3.25. Dialogue completion is not a universal compromise (14/20 shortcuts; 3/20 clear KC). Critics disagree on KC attribution for 14/20 opportunities and platform plausibility for 12/20. Greater conversational naturalness can therefore weaken the scorable opportunity boundary KT requires.

6.4 Supporting linguistic generalisation

The original known-truth benchmark still answers a question that real logs cannot score directly. K^* loss is .669161 on seen grammar, .672036 on 15 pairwise-seen/full-tuple-unseen cells, and .681181 on six unseen-value cells; exact-cell histories cost .037609/.028627 on the two held-out regimes. This is narrow compositional extrapolation under a generator whose KCs all have seen support. It is not unrestricted human grammatical transfer, and the measurement audit prevents treating this clean result as a platform finding.

Downstream hypothesis	KCs	Exact K^* KCs	Activation Jaccard	Aligned edge F1	Probe loss vs. ceiling
Compositional ceiling	18	18/18	1.000000	1.000000	.669606 (reference)
Selected atomic class	18	16/18	.970854	.965385	+.000374 [.000228, .000517]
RQ3 operation groups	8	5/18	.371913	.750929	+.005864
RQ3 seen-cell fine	54	1/18	.084184	.186969	+.006657
Hash distractors	18	0/18	.202342	.359259	+.013238

Table 6: RQ3 structural and predictive recovery after observable-only selection is frozen. Jaccard is padded optimal activation-column matching over the complete bank. The compositional row is a positive-control ceiling, not a blindly selected policy; only the selected-versus-ceiling delta is shown with its preregistered learner interval. The 8- and 54-KC controls are discovery-protocol baselines, distinct from the RQ2 6- and 75-KC conditions.

Error history	LL	Loc.@1	KC-evid.	RMSE
Binary only	.636359	.420781		.228727
Linked positive control	.635493	1.000000		.144357
Linked on 80%	.635833	.883728		.158804
Within-item shuf-fled	.636987	.462525		.165519

(a) Structured-error controls; three-seed means.

Format	Det.	KC clear	Shortcut	Response LB
Constrained cloze	17/20	17/20	1/20	1.30
Dialogue completion	0/20	3/20	14/20	3.85
Open dialogue	0/20	4/20	13/20	4.55

(b) Automated dialogue-continuum judgments.

Assignment	D LL	Δ LL vs. lab	CI away	State RMSE	Item Gini	Median gap
Q-balanced lab	.636359	—	—	.129724	.162530	93.67
Curriculum	.639779	+.003420	2/3	.139472	.162530	31.00
Mixed practice	.636341	−.000018	0/3	.130748	.162530	92.00
Adaptive cell weakness	.639477	+.003118	1/3	.141659	.080298	26.67

(c) Post-response, frozen-before-fit exploratory policy recovery.

Table 7: Response, dialogue, and assignment boundaries. (a) targets a synthetic post-outcome attribution; KC-evidence RMSE is a secondary count diagnostic. (b) reports model judgments, not human ratings. (c) uses item-prerequisite state RMSE; lab, curriculum, and mixed share an occurrence multiset and terminal oracle state, so this is recovery, not policy efficacy.

7 Discussion

Known truth is necessary, not sufficient. Full-v1 supplies the counterfactual quantities that ordinary learner logs do not: exact opportunity boundaries, generator KCs, Q, mastery trajectories, and paired response worlds. That makes representation errors measurable. It does not make a stored prompt understandable, its accepted-answer set fair, or its response process human-like. The audits and failed matched-bank construction are therefore part of the scientific result, not ancillary quality control. They show that artifact integrity, Q geometry, and measurement validity answer different questions.

RQ1: construction requires explicit validity layers. The grammar-to-cell stage remains useful because it preserves scope and uncertainty without turning every descriptor into a KC. The next two stages are less settled. Independent KC proposals agree on only nine activation signatures, while item critics disagree by role on half the bank. A nominally simple item may instantiate its cell yet remain underdetermined, demand an undeclared

operation, or encode an annotation protocol instead of a learner interface. Conversely, a natural-looking task can provide weak evidence about the target. A credible construction pipeline must therefore gate linguistic fidelity, answerability, KC purity, and deployability separately; no average “realism” score or full-rank Q can compensate for a critical failure.

RQ2: nuisance can masquerade as skill structure. In the clean control, unnecessary KC splits lose to shared K^* . After a format effect is planted, the format-specific split improves because it can absorb measurement variance, even though learner mastery remains shared. An observable format term performs better in that strong control. This is direct counterfactual evidence for a familiar interpretive risk: predictive support for a finer KC ontology need not imply distinct learnable competencies. However, the correction is scenario-dependent. It is not consistent in the combined heterogeneous world, and the item-only experiment does not reproduce the legacy split reversal. Model D succeeds because the planted effect lies in its same-seen-

Representation	Seen (54 cells)	Unseen combination (15)	Unseen value (6)
Generator K^* (absolute LL)	.669161	.672036	.681181
RQ3 atomic (delta)	≈ 0	≈ 0	-.003236 (n.s.)
Family union (delta)	+.008940 [†]	+.009185 [†]	-.001751 (n.s.)
Structural split-2 (delta)	+.002234 [†]	+.008806 [†]	-.000684 (n.s.)
Exact cell (delta)	+.008209 [†]	+.037609 [†]	+.028627 [†]
K^* + intersections (delta)	+.001415 [†]	+.001196 [†]	+.006954 [†]

Table 8: RQ4 event-weighted log loss by semantic grammar regime. Deltas are candidate minus K^* under one common fitting protocol. A dagger marks a 2,000-repeat whole-learner interval strictly above zero; n.s. marks an interval spanning zero. All unseen combinations are pairwise-seen and full-tuple-unseen.

item basis by construction. It is an oracle-aligned positive control, not a solution for new items or arbitrary real-platform difficulty.

RQ3: information design sets the inferential ceiling. Activation-equivalent ontologies cannot be separated by more responses on the same Q rows. Anchor contrasts can restore structural distinguishability, but weak effects can remain practically hard to detect. Structured error labels can localise a planted failed KC while barely changing response prediction; the shuffled control further shows that a convenient state diagnostic can improve for the wrong reason. Assignment policies select different histories, so recovery depends on the platform process as well as the learner. Finally, the dialogue continuum exposes the same tension at the item surface: open interaction looks more conversational while multiplying reasonable responses, incidental grammar, shortcuts, and attribution uncertainty. Ecological plausibility and measurement precision are axes to report, not endpoints of a single scale.

Implications for real data collection. The results motivate crossed KCs and formats, multiple semantic variants, diagnostically distinct anchors, stable item identifiers, explicit response mechanisms and accepted-answer policies, raw learner responses, error and context metadata, and the assignment policy or propensity that produced each opportunity. Item and format effects should be modelled before interpreting a KC split as learner structure. Non-updating probes can help separate measurement from practice where ethically and operationally feasible. These are design principles and failure-mode checks; synthetic effect sizes and sample thresholds are not literal platform prescriptions.

What should be validated next. The declared human/expert protocol separates six targets: GrammarCell fidelity, KC pedagogical plausibility, item

deployability, learner answerability, format plausibility, and error realism. It uses rendered tasks, role-separated expert review, and a balanced learner solvability study. Such evidence is necessary before deploying a successor bank. The failed automated campaign is useful because it identifies where declared correction and revalidation are needed; it is not a rationale to lower the release gate.

Language-independence boundary. Projection, Q auditing, simulation, perturbation, and evaluation are schema-driven, and a non-English-named toy schema exercises their interfaces. EGP parsing, the six dimensions, English operation declarations, prompts, and error interpretations remain language-specific. This is reusable methodology, not cross-lingual empirical validity or a universal grammar ontology.

8 Reproducibility and Audit Trail

The immutable reference is rooted at data/grammar_kt_full_v1/. Its manifest records the pre-response revision, exact simulator command, configurations, inputs, deterministic gzip metadata, row counts, and an 88-file recursive inventory. Independent replay reconstructs the 113×18 Q and all 283,000 linked public/private events. The measurement programme stores a separate baseline anchor containing the protected tree and core hashes; no extension command writes inside full-v1.

Every LLM-backed scientific stage freezes its prompt, exact input, provider schema, model and reasoning setting, raw response, parsed response, and hashes. The complete item audit, three KC inductions, matched-bank attempts, and dialogue critics retain role-specific output rather than a collapsed score. Infrastructure failures are labelled separately from scientific failures. In particular, the matched-bank negative result is replayed from its immutable candidate, solver, critic, and curation

evidence.

Deterministic scripts under scripts/ experiments/ reconstruct the cross-audit synthesis, bank funnel, controlled worlds, structured-error comparisons, and exploratory policy recovery. The controlled-scenario study plan hashes the structural instrument and implementation before responses. Observable archives exclude mastery, Q, response probabilities, and planted failure attributions; oracle archives are opened only by declared evaluations. Cross-world contrasts verify learner sets and exact evaluation-row identity.

Because the controlled response archives contain 4.59 million events, the programme-level evidence package retains compact aggregates, manifests, exact hash contracts, and reconstruction commands rather than treating generated rows as a new released dataset. The final allow-list manifest also excludes failed-attempt scratch trees, caches, and incomplete staging artifacts. The executable measurement notebook reads only retained compact evidence, and the paper build, notebook execution, full test suite, and full-v1 replays are part of final verification.

9 Limitations and Ethics

No human measurement validation. Both full-bank audits, the KC inductions, the matched-bank critics, and the dialogue pilot are model-based stress tests. No learner, teacher, psychometrician, or platform reviewer assessed the rendered tasks. Role disagreement is informative but is not interrater reliability, and plurality is not validity. CEFR appropriateness, accessibility, bias, scorer coverage, and actual response processes remain unmeasured. Accordingly there is **no new dataset release**; full-v1 remains the clean reference and none of the extension artifacts is classroom-ready or release-valid.

Item surface and matched-bank failure. Full-v1 learners never read prompts or produce text: item IDs select Q rows and binary outcomes are sampled from mastery. The 113-item audit therefore cannot change its responses. The intended 152-slot matched bank failed its preregistered complete-family, coverage, and rank gates. The 20 passing slots are not a representative partial bank. A future declared-correction layer would need independent replay and then rendered human/expert and learner validation before any deployment or release claim.

Content-free controlled scenario. The quantitative format/item study uses structural slots and categorical format labels, not learner-facing content. Planted effects are sensitivity levels rather than estimates from human data. Orthogonality holds over equally weighted seen slots, not exactly over the acquisition event multiset. Model D is aligned to the planted same-seen-item residual basis; held-out item effects are encoded as zero. Its advantage does not demonstrate arbitrary item deconfounding or unseen-item generalisation. Explicit format adjustment is not consistently superior to the false split after learner heterogeneity is added.

Errors and learner state. The structured failed-KC label is sampled after an incorrect outcome in proportion to active mastery deficit. It is neither causal truth nor a validated learner-error taxonomy; the 80% condition hides labels rather than misclassifying them. The Beta evidence-count terminal-KC estimate is not a fitted mastery trajectory, and its improvement under shuffled labels exposes metric bias. We therefore do not scale to synthetic error text or claim that error-aware KT recovers human knowledge.

Platform schedules. Q-balanced, curriculum, and mixed policies share one occurrence multiset and, under order-independent learning, identical terminal oracle mastery. Their fitted differences concern history representation, not learning efficacy. Adaptive assignment changes exposure and state; its logged propensity is tied to the deterministic policy and is not a general off-policy-evaluation weight. The alternative-policy fits were frozen after responses and are exploratory.

Statistical uncertainty. Learner bootstrap intervals condition on one fitted model, frozen train/dev sample, bank, seed, hyperparameter, and simulator world. They omit refitting, item/cell sampling, bank construction, simulator-family and seed-level uncertainty, multiplicity, and any human population. Three-seed ranges are descriptive. Fitted terminal learner-KC recovery was not implemented for the new A–D models; item-prerequisite state and evidence-count diagnostics must not be relabelled as that estimand.

Linguistic and generalisation scope. The empirical domain is English single-main-clause verbal morphosyntax under six dimensions. Source normalisation is model-assisted and has no expert adjudication or second-language replication. The six

unseen-value cells all concern perfect-progressive aspect, and every generator KC has seen support. Neither the alternate-schema software test nor this narrow probe establishes cross-lingual or unrestricted grammatical generalisation.

Ethics. No human-participant or learner records are used; all response histories are synthetic. Latent conditions are modelling assumptions, not demographic or ability groups. Generated exercises and error forms must not be used for assessment or instruction without qualified review for correctness, accessibility, fairness, and educational value. The consult-only source is not redistributed, and its use remains subject to its licence.

10 Conclusion

Synthetic perfect-information data can answer questions that real language-learning logs usually cannot: whether a supplied KC system matches the generator, how Q errors alter inference, and how well fitted state tracks known mastery. Full-v1 preserves those clean controls. This programme shows why known truth alone is not enough. Learner-facing answerability, KC interpretation, item and format nuisance, assignment history, and response information all constrain what KT results mean.

The strongest evidence is deliberately bounded. A planted format effect can make false format-specific KCs look predictive, explicit nuisance modelling can recover the shared structure in that control, structured errors can localise a planted failure without greatly improving prediction, and open dialogue trades answer determinacy for visible naturalness. The matched platform-oriented bank did not pass its release gate, so we preserve the negative result and make no new dataset release. The contribution is thus an auditable controlled analysis of when grammar KT is informative or misleading, together with concrete data-collection requirements for testing those conclusions with real learners.

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A Artifact Map

Scientific object	Inspectable declaration and frozen evidence
Source and GRAMMAR-CELLS	modules/grammar/resource/egg/; modules/grammar/canonical/; data/grammar_kt_full_v1/grammar/; normalisation and inventory provenance
Generator K^*	modules/kcs/generator/; data/grammar_kt_full_v1/kcs.jsonl; construction provenance
Item bank	modules/items/; data/grammar_kt_full_v1/items/items.jsonl; immutable generation, validation, correction, and curation evidence
True Q^*	scripts/build_true_q_matrix.py; data/grammar_kt_full_v1/q_matrix.csv; data/grammar_kt_full_v1/oracle/q_matrix_sparse.jsonl; data/grammar_kt_full_v1/provenance/measurement/audit.json
Observable and oracle events	modules/simulation/; scripts/freeze_baseline_dataset.py; data/grammar_kt_full_v1/interactions.jsonl.gz; data/grammar_kt_full_v1/oracle/learner_truth.jsonl.gz
Downstream hypotheses	scripts/experiments/; experiments/full_v1/; reports/full_v1_artifacts/
Measurement audits	experiments/measurement_realism/audits/; reports/platform_plausibility_audit.md; reports/kc_methodology_audit.md
Matched-bank evidence	experiments/measurement_realism/design/bank_protocol/; immutable generation, solver, critic, and failed-release-gate evidence
Controlled scenarios	experiments/measurement_realism/worlds/controlled_instrument_v1/; compact nuisance, error, and policy analyses with non-release manifests
Release and claims	data/grammar_kt_full_v1/manifest.json; reports/final_release_manifest.json; reports/measurement_extension_rq_ledger.md; ACL/evidence.md

Table 9: Repository-root map from scientific objects to their inspectable declarations and frozen evidence.

B Generator and Measurement Detail

The 18 generator KCs fall into five auditable groups:

- finite-form selection: present and past;

- aspectual dependencies: perfect and progressive;
- voice/polarity: BE-passive and verbal negation;
- clause operations: imperative, polar question, and non-subject-WH; and
- central-modal identity: can, could, may, might, must, shall, should, will, and would.

Seventeen recur across cells. Seven have fewer than six final items, but every KC has at least two items and 12 seen acquisition opportunities per learner. The perfect-progressive-chain alternative would be active in six cells but is nested in both aspect dependencies; it was excluded before item generation.

The final bank contains 84 seen, 20 unseen-combination, and nine unseen-value items. Max-one preserves all 75 cells with 75 items, while max-three adds only 13 items beyond max-two. No selected prompts are exact duplicates. Surface nonidentity is not itself evidence of pedagogical diversity.

C Research-Question Ledger

RQ1—auditability supported; platform validity not established. Full-v1 is complete and replayable, but only 60/113 items are usable in both automated audits, and the matched-format successor freezes only 5/38 families. There is no new dataset release.

RQ2—supported for declared controls. Coarsening, refinement, and Q corruption harm prediction in clean full-v1. In the separate controlled scenario, planted format nuisance increases a false split’s relative advantage and an observed format term repairs the strong control. The correction is inconsistent under combined heterogeneity; the aligned item basis is only a positive control.

RQ3—unique recovery rejected; information-design effects supported. Equivalent seen-Q rules remain observationally indistinguishable, while anchors restore selected contrasts. Linked structured errors improve planted failure localisation more than prediction; schedule changes alter model recovery; open dialogue improves naturalness while weakening determinacy and KC attribution.

The original grammar-generalisation result remains supporting evidence: reusable KCs outperform exact-cell histories on pairwise-seen/full-tuple-unseen cells, within a narrow synthetic regime. Every answer is conditional on the declared generator, instrument, assignment policy, and response-information world.

Audit case	Learner-visible task (abridged only where marked)	Why the judgment concerns measurement
Shared clear pass	“By lunchtime, the room _____. (paint)” after stating that workers finished before lunch.	The temporal contrast and lexical cue select <i>had been painted</i> ; both audits accept the perfect–passive–past cloze.
Shared answer-space failure	“The workers were gone and the gate was open. The gate ____.” (cue: open).	<i>was open</i> and <i>had been open</i> are salient alternatives to keyed <i>had been opened</i> ; failure cannot isolate the intended Q row.
KC-purity disagreement	“Mia said, ‘I will carry one bag.’ Mia said that she ____.”	The one-KC row names WOULD, but the task also invokes reported-speech backshift and pronoun change; the strict and role-based audits disagree.
Modal answer-space failure	“The sky is dark, so take an umbrella. It ____ soon.” (cue: rain).	<i>may</i> , <i>might</i> , <i>will</i> , and <i>is going to</i> are plausible, so the context cannot uniquely measure the planted MAY KC.

Table 10: Exact full-v1 audit cases (prompt text lightly abridged where indicated). The examples separate grammatical instantiation from answer determinacy and KC measurement purity; model-critic judgments are not human validation.